

A Packing of 14 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 14 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.392880. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.380617, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 14 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.392880 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 14$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.380617, obtained by deleting 1 of the 15 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 14/13 = 2.393162$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N14.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
797626779713416869857072463894110760216776703040320886722119224483970052
708456577863607561585947508378822971774418122747936850323927220087376234
442918977766638599456142780782397326762829669365
```

denominator:

```
333333333333795980716948609937496416884568507631556217129981526167045865
035589570946959042668531675651584674070506794436493865768347210391466343
693823441912530744453846213765836839691025002152
```

B Coordinates

14 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N14.txt`.

```
325398008346 -54702146338 -692843984129 19555166337617430021660 -183900931257 -333264467153 -314413055082225214374 158598251064 271350690624 -403358665112
564782217841 447950296081 75420339725 137612766211 -58432436648 55016014407 -360332231243 479030379021-219827776373 -210628269052 83782700746 592368959956
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